







Python Environment Setup Guide

Pablo Tassi

This document provides step-by-step instructions for setting up a Python environment suitable for executing any Jupyter notebook distributed as part of the project, on both 🐧 **Linux** and ⊞ **Windows**.

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Projet Energies Marines

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1 Overview

The notebook relies on a small, well-defined set of scientific Python packages plus the **Copernicus Marine Toolbox**, used to download ocean current and temperature data directly from the Copernicus Marine Service catalogue.

Package	Purpose	Notes
`numpy`	Numerical arrays & math	Core scientific computing
`xarray`	NetCDF / labeled N-D arrays	Reads the downloaded ocean datasets
`matplotlib`	Plotting	Figures, maps, colour bars
`cartopy`	Map projections & coastlines	Has compiled C dependencies (GEOS, PROJ)
`copernicusmarine`	Copernicus Marine Toolbox	Login & `subset()` data download
`jupyterlab` / `ipykernel`	Notebook runtime	To run/open the `.ipynb` file

Recommended approach (for Windows)

We strongly recommend using **conda** (via Miniconda) rather than plain `pip`/`venv`. `cartopy` depends on compiled geospatial libraries (GEOS, PROJ) that conda-forge ships as pre-built binaries – this avoids painful compilation errors, particularly on Windows.

2 Prerequisites

- A free **Copernicus Marine Service** account – register at `https://data.marine.copernicus.eu` (needed to download the ocean data).
- About **2 GB** of free disk space (conda environment + downloaded data).
- A terminal (bash/zsh) – **or** the *Anaconda Prompt / PowerShell*.
- (Optional but recommended) Visual Studio Code with the *Python* and *Jupyter* extensions, to open the notebook.

3 Step-by-Step Setup

The steps below are **identical in structure** for both operating systems – only the terminal application and a few command-line details differ.

Step 1 – Install Miniconda

Linux – Step 1: Install Miniconda

bash

```
# Download the Miniconda installer
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

# Run the installer (follow the on-screen prompts)
bash Miniconda3-latest-Linux-x86_64.sh

# Reload your shell so that "conda" is recognized
source ~/.bashrc
```

Windows – Step 1: Install Miniconda

- Download the installer from https://docs.conda.io/en/latest/miniconda.html (choose *Miniconda3 Windows 64-bit*).
- Double-click the `.exe` file and follow the installation wizard (default options are fine).
- From the **Start Menu**, open **"Anaconda Prompt (miniconda3)"** – all commands below are typed there.

Step 2 – Create a dedicated environment

Creating an isolated environment keeps this tutorial's dependencies separate from any other Python project on your machine. The command is the same on both systems.

Terminal / Anaconda Prompt

```
conda create -n copernicus-notebook -c conda-forge \
    python=3.11 numpy xarray matplotlib cartopy jupyterlab ipykernel -y
```

What this does

Creates a new environment named `copernicus-notebook`, using the `conda-forge` channel (which provides up-to-date, pre-compiled packages, including `cartopy`'s geospatial dependencies).

Step 3 – Activate the environment

Linux – Step 3: Activate

bash

```
conda activate copernicus-notebook
```

Windows – Step 3: Activate

Anaconda Prompt

```
conda activate copernicus-notebook
```

Same command, both systems

Conda environment activation uses the exact same syntax on Linux and Windows. Your prompt should now show `(copernicus-notebook)` at the beginning of the line.

Step 4 – Install the Copernicus Marine Toolbox

The Toolbox is not (yet) available on conda-forge for all platforms, so it is installed via `pip` *inside* the already-activated conda environment.

Terminal / Anaconda Prompt (env. active)

```
pip install copernicusmarine
```

Step 5 – Register the Jupyter kernel

This makes the `copernicus-notebook` environment selectable as a kernel inside Jupyter Notebook, JupyterLab, or VS Code.

Terminal / Anaconda Prompt (env. active)

```
python -m ipykernel install --user --name copernicus-notebook \
    --display-name "Copernicus notebook (Python 3.11)"
```

Step 6 – Launch and open the notebook

Linux – Step 6: Launch

bash

```
cd path/to/Notebook_Python_Tutorial
jupyter lab
```

Windows – Step 6: Launch

Anaconda Prompt

```
cd path\to\Notebook_Python_Tutorial
jupyter lab
```

Using VS Code instead

If you prefer VS Code: open the tutorial folder, open the `.ipynb` file, click **"Select Kernel"** (top right) and choose **"Copernicus notebook (Python 3.11)"**.

4 One-shot install (copy/paste)

For convenience, here are all commands combined into a single block for each operating system.

Linux – All-in-one

bash

```
conda create -n copernicus-notebook -c conda-forge \
    python=3.11 numpy xarray matplotlib cartopy jupyterlab ipykernel -y
conda activate copernicus-notebook
pip install copernicusmarine
python -m ipykernel install --user --name copernicus-notebook \
    --display-name "Copernicus notebook (Python 3.11)"
jupyter lab
```

Windows – All-in-one

Anaconda Prompt

```
conda create -n copernicus-notebook -c conda-forge ^
    python=3.11 numpy xarray matplotlib cartopy jupyterlab ipykernel -y
conda activate copernicus-notebook
pip install copernicusmarine
python -m ipykernel install --user --name copernicus-notebook ^
    --display-name "Copernicus notebook (Python 3.11)"
jupyter lab
```

Windows line continuation

On Windows, the line-continuation character in `cmd.exe` / Anaconda Prompt is the caret ^ (not the backslash \). If you use PowerShell instead, replace ^ with the backtick `, or simply type each command on a single line.

5 Copernicus Marine login

Before downloading any data, the notebook calls `copernicusmarine.login()`. The first time you run it, it will prompt you for your Copernicus Marine username and password (from your free account) and store them locally, so you won't need to log in again on the same machine.

Free registration

No account yet? Register for free at `https://data.marine.copernicus.eu` – it only takes a couple of minutes.

6 Alternative: pip + venv

Important

Installing `cartopy` via plain `pip` requires the GEOS and PROJ system libraries to already be present, which is straightforward on Linux but notoriously difficult on Windows. Use this route only if you cannot install conda.

Linux – pip + venv

bash

```
sudo apt-get update
sudo apt-get install -y libgeos-dev libproj-dev proj-data proj-bin

python3 -m venv copernicus-notebook-venv
source copernicus-notebook-venv/bin/activate

pip install --upgrade pip
pip install numpy xarray matplotlib cartopy jupyterlab ipykernel copernicusmarine

python -m ipykernel install --user --name copernicus-notebook-venv \
    --display-name "Copernicus notebook (venv)"
```

Windows – pip + venv

PowerShell

```
py -3.11 -m venv copernicus-notebook-venv
.\copernicus-notebook-venv\Scripts\Activate.ps1

pip install --upgrade pip
pip install numpy xarray matplotlib cartopy jupyterlab ipykernel copernicusmarine

python -m ipykernel install --user --name copernicus-notebook-venv `
    --display-name "Copernicus notebook (venv)"
```

If `cartopy` fails to build, install the pre-compiled wheel instead: `pip install cartopy -only-binary :all:`, or switch to the conda route described in Section 3.

7 Verifying the installation

Run the following snippet in a notebook cell (or a plain `.py` script) to confirm that every package is correctly installed and importable.

`verify_env.py`

```
import cartopy
import copernicusmarine
import matplotlib
import numpy
import xarray

print("numpy :", numpy.__version__)
print("xarray :", xarray.__version__)
print("matplotlib :", matplotlib.__version__)
print("cartopy :", cartopy.__version__)
print("copernicusmarine:", copernicusmarine.__version__)
print("\nAll packages imported successfully!")
```

Expected result

No errors, and a short version report is printed for each package. If any import fails, jump to Section 8 for troubleshooting tips.

8 Troubleshooting

Symptom	Likely fix
`cartopy` fails to build / `ImportError: libgeos_c`	Use the conda-forge route (Section 3). It ships pre-built GEOS/PROJ binaries.
`conda: command not found` (Linux)	Re-open the terminal, or run `source ~/.bashrc` / `source ~/miniconda3/etc/profile.d/conda.sh`.
Kernel not listed in Jupyter/VS Code	Re-run the `ipykernel install` command from Step 5, then restart Jupyter/VS Code.
`copernicusmarine.login()` hangs or fails	Check your internet connection / proxy settings; verify your account credentials at https://data.marine.copernicus.eu.
SSL certificate errors on a corporate network	Configure the `HTTPS_PROXY` / `REQUESTS_CA_BUNDLE` environment variables as required by your organisation's network policy.
Very slow `conda create`	Use `conda config -set channel_priority strict` once, or install `mamba` (`conda install -n base -c conda-forge mamba`) and replace `conda` with `mamba` in Step 2.

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